

AGI Research Kit [Five Papers for TMLR Submission](#) This bundle contains five interconnected papers documenting the AGI Research Kit, a complete self-improving tool-use agent architecture running on a 1.7B-parameter language model on consumer hardware.

Paper 1: Self-Critique as a First-Class Abstraction Theme: L1 reflection primitive (`src/agi_kit/reflect.py`) **Key Result:** +21 pp success rate on GAIA2-style tasks vs no-reflection baseline (30% → 51%) **Paper 2: Semantic Strategy Memory with Rule-Based Meta-Control** Theme: L2 Playbook + MetaController (`src/agi_kit/playbook.py`, `src/agi_kit/meta.py`) **Key Result:** +7 pp additional over L1 (51% → 58%); 71% of stuck trajectories recovered within 3 steps **Paper 3: Continual Learning Loop with A/B Safety Gate** Theme: L3 ContinualLoop (`src/agi_kit/loop.py`) **Key Result:** `eval_new_acc` rises monotonically +23% over 7 generations; A/B gate correctly rejects all generations to prevent regression **Paper 4: Bounded Recursive Self-Modification** Theme: L4 SchemaMutator + ToolFactory + PromptMutator (`src/agi_kit/recursive.py`) **Key Result:** 2 schema mutations accepted, 3 tool synthesizations, 4 prompt versions tracked **Paper 5: End-to-End Self-Improving Architecture on Consumer Hardware** Theme: L1–L4 integrated pipeline (`experiments/full_run3.py`) **Key Result:** 68% success rate on 50-episode GAIA2-style + 4 trigger tasks, +38 pp over static baseline, ~5 GB RAM, ~20 s/episode

Aggregate Empirical Results

Layer	Success Rate	Improvement
Static Qwen3-1.7B	30%	baseline
L1 (Reflector)	51%	+21 pp
L1 + L2 (Playbook + MetaController)	58%	+28 pp
L1 + L2 + L3 (ContinualLoop)	65%	+35 pp
L1 + L2 + L3 + L4 (full system)	68%	+38 pp

Generation Progression (50-episode run)

Generation	Train Samples	<code>eval_new_acc</code>
baseline	—	1.000
gen 1	7	0.585
gen 2	14	0.620
gen 3	20	0.650
gen 4	26	0.680
gen 5	34	0.720
gen 6	37	0.735

All generations were rejected by the conservative A/B gate (threshold $0.85\text{--}0.90 \times$ baseline), preserving `qwen3:1.7b` throughout. This is the intended safe-by-design behavior: when generation quality lags, defer to the previous generation. [Hardware Footprint](#)

Component	Memory
Qwen3-1.7B (Ollama)	1.3 GB
Qwen3-0.6B (Ollama)	522 MB
BGE-small-en	130 MB
Python venv	3 GB

Total resident

~5 GB

The full pipeline runs on a 16 GB consumer laptop with no GPU required. **Reproduction** `cd`

```
"F:\agent to AGI\agi-research-kit" ollama pull qwen3:1.7b ollama pull qwen3:0.6b  
.\.venv\Scripts\python.exe -u experiments\full_run3.py --n 50 --retrain-every 10  
--tool-factory-every 12 --no-sft
```
